

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Analyse and Practise

Name: _____

Date: _____

LEARNING INTENTION	pose testable questions, predict and plan safe repeatable investigations
KEY IDEA	A testable question names what changes, what is measured and what is controlled. A plan includes repeat trials, risk controls, permissions and an evidence-based prediction.
WORKED MODEL	Use the cycle: question -> prediction -> safe method -> precise data -> representation -> evidence-based conclusion -> evaluation -> communication.

SUCCESS CRITERIA - I CAN

[] apply the inquiry skill [] use evidence [] evaluate quality [] explain a next step

A. NOTICE AND IDENTIFY

1. Turn a broad topic into a testable question with changed and measured variables.

2. Identify independent, dependent and three controlled variables.

3. Distinguish prediction from guess by adding scientific reasoning.

B. APPLY AND EXPLAIN

4. Write a repeatable step sequence for the scenario.

5. Complete a risk table: hazard, possible harm, control and response.

6. Explain required permissions and respectful planning for work on Country/Place where relevant.

C. INDEPENDENT FLUENCY - CHECK AND CREATE

7. Design a results table before collecting data.

8. Audit the plan for fairness, repetition, precision and safety.

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D. APPLY THE SKILL IN CONTEXT

9. Scenario: test how ramp surface affects a toy vehicle's travel distance Write the investigable question and reasoned prediction.

10. Specify variable ranges, equipment, units, controls and number of trials.

11. Explain how the plan would allow another group to repeat the investigation.

E. REASON, EVALUATE AND TRANSFER

12. Diagnose a plan that changes two variables and measures once.

13. Revise the design and justify three improvements.

F. STUDENT REVIEW - PROVE YOUR UNDERSTANDING

[] I can pose testable questions, predict and plan safe repeatable investigations.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult decision was _____ because _____

One correction or improvement I made: _____

My next practice step: _____

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Accept equivalent responses when the student applies the targeted skill accurately and justifies the choice.

COMPLETE ANSWER KEY

1. Question must be measurable and name a relationship.
2. Independent=changed; dependent=measured; controls=kept consistent.
3. Prediction needs a directional outcome and science reason.
4. Steps must be numbered, precise and repeatable.
5. Controls must reduce each identified risk.
6. Include access, cultural and environmental permissions where applicable.
7. Table needs variable values, trials, units and summary column.
8. Check one changed variable, repeated measures and safe precision.
- 9-11. Accept a coherent scenario-specific fair-test plan.
12. Cause cannot be isolated and reliability cannot be judged.
13. Change one variable, repeat trials, standardise measurement.

TEACHER CHECK AND NEXT STEP

[] Uses accurate science ideas and evidence.

[] Explains a decision rather than listing an answer.

Next step: model one difficult item, complete one together, then transfer the skill to a new investigation.

COMMON MISCONCEPTION / WATCH OUT

A topic is not an investigable question; it must be testable with measurable variables.

A fair test changes one variable while relevant conditions remain controlled.

EXTENSION AND RE-TEACH

Re-teach: model one item aloud, complete one together, then ask the student to explain an independent correction.

Extension: transfer the skill to a short authentic text and justify two deliberate language choices with quoted evidence.